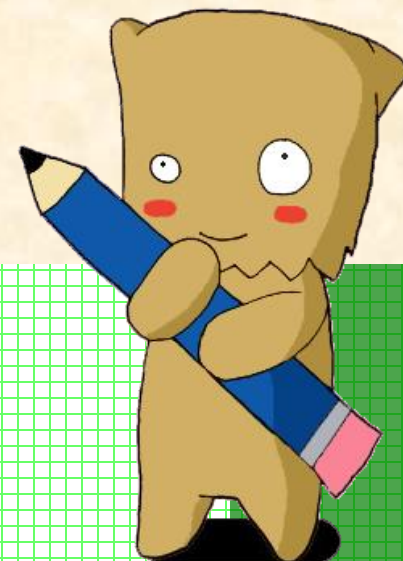




Chapter 5

DECISION MAKING FOR TWO SAMPLES



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Decision Making for Two Samples



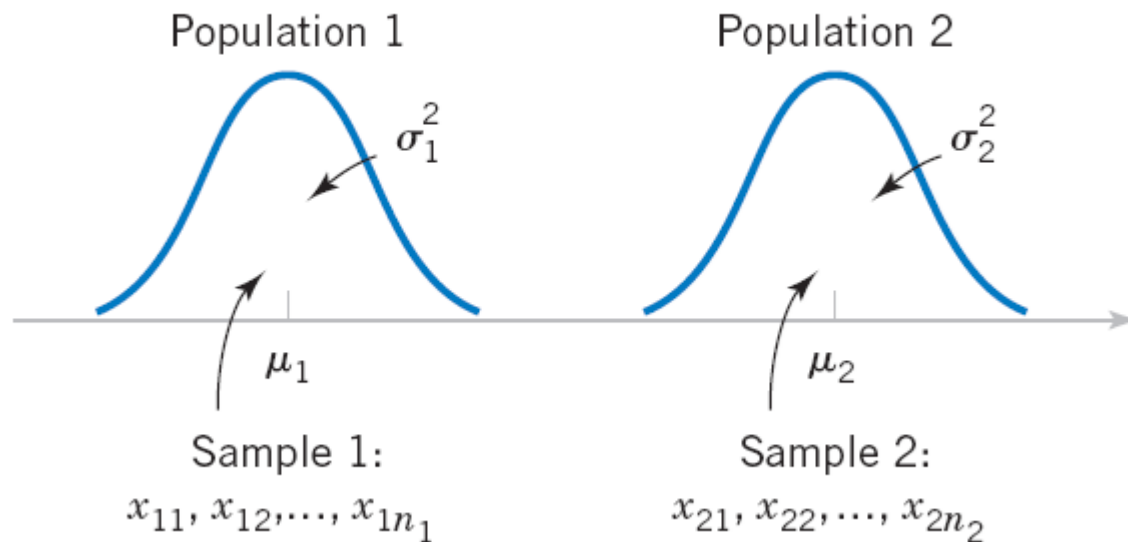
- Inference on the Mean of Two Population, Variances Known
- Inference on the Mean of Two Population, Variances Unknown
- The Paired t-Test
- Inference on the Ratio of Variances of Two Normal Populations
- Inference on Two Population Proportions

Introduction

- The previous chapter presented hypothesis tests and confidence intervals for a single population parameter (the mean μ , the variance σ^2 , or a proportion p).
- This chapter extends those results to the case of two independent populations.
- Most of the practical applications of the procedures in this chapter arise in the context of simple comparative experiments in which the objective is to study the difference in the parameters of the two populations.



Inference on the Mean of Two Population, Variances Known



Two independent populations.

Inference on the Mean of Two Population, Variances Known

Assumptions

1. $X_{11}, X_{12}, \dots, X_{1n_1}$ is a random sample from population 1.
2. $X_{21}, X_{22}, \dots, X_{2n_2}$ is a random sample from population 2.
3. The two populations represented by X_1 and X_2 are independent.
4. Both populations are normal.

$$E(\bar{X}_1 - \bar{X}_2) = E(\bar{X}_1) - E(\bar{X}_2) = \mu_1 - \mu_2$$

$$V(\bar{X}_1 - \bar{X}_2) = V(\bar{X}_1) + V(\bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

Inference on the Mean of Two Population, Variances Known

The quantity

$$Z = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

has a $N(0, 1)$ distribution.



Inference on the Mean of Two Population, Variances Known

- Hypothesis Tests for a Difference in Means, Variances Known

Null hypothesis: $H_0: \mu_1 - \mu_2 = \Delta_0$

Test statistic:
$$Z_0 = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Alternative Hypotheses	P-Value	Rejection Criterion For for Fixed-Level Tests
$H_1: \mu_1 - \mu_2 \neq \Delta_0$	Probability above $ z_0 $ and probability below $- z_0 $, $P = 2[1 - \Phi(z_0)]$	$z_0 > z_{\alpha/2}$ or $z_0 < -z_{\alpha/2}$
$H_1: \mu_1 - \mu_2 > \Delta_0$	Probability above z_0 , $P = 1 - \Phi(z_0)$	$z_0 > z_\alpha$
$H_1: \mu_1 - \mu_2 < \Delta_0$	Probability below z_0 , $P = \Phi(z_0)$	$z_0 < -z_\alpha$

Example: Paint Drying Time

A product developer is interested in reducing the drying time of a primer paint. Two formulations of the paint are tested; formulation 1 is the standard chemistry, and formulation 2 has a new drying ingredient that should reduce the drying time. From experience, it is known that the standard deviation of drying time is 8 minutes, and this inherent variability should be unaffected by the addition of the new ingredient. Ten specimens are painted with formulation 1, and another 10 specimens are painted with formulation 2; the 20 specimens are painted in random order. The two sample average drying times are $\bar{x}_1 = 121$ minutes and $\bar{x}_2 = 112$ minutes, respectively. What conclusions can the product developer draw about the effectiveness of the new ingredient, using $\alpha = 0.05$?



Example: Paint Drying Time

1. **Parameter of interest:** The quantity of interest is the difference in mean drying times, $\mu_1 - \mu_2$, and $\Delta_0 = 0$.
2. **Non hypothesis:** $H_0: \mu_1 - \mu_2 = 0$, or $H_0: \mu_1 = \mu_2$.
3. **Alternative hypothesis:** $H_1: \mu_1 > \mu_2$. We want to reject H_0 if the new ingredient reduces mean drying time.
4. **Test statistic:** The test statistic is

$$z_0 = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

where $\sigma_1^2 = \sigma_2^2 = (8)^2 = 64$ and $n_1 = n_2 = 10$.

5. **Reject H_0 if:** Reject $H_0: \mu_1 = \mu_2$ if the P -value is less than 0.05.
6. **Computations:** Since $\bar{x}_1 = 121$ minutes and $\bar{x}_2 = 112$ minutes, the test statistic is

$$z_0 = \frac{121 - 112}{\sqrt{\frac{(8)^2}{10} + \frac{(8)^2}{10}}} = 2.52$$

7. **Conclusion:** Since $z_0 = 2.52$, the P -value is $P = 1 - \Phi(2.52) = 0.0059$, so we reject H_0 at the $\alpha = 0.05$ level

Practical Interpretation: We conclude that adding the new ingredient to the paint significantly reduces the drying time. This is a strong conclusion.



Exercise – Wax

- Independent groups, population standard deviations known. The mean lasting time of two competing floor waxes is to be compared. Twenty floors are randomly assigned to test each wax. Both populations have a normal distributions. The data are recorded in the table.

Wax	Sample Mean	Population Standard Deviation
1	3	0.33
2	2.9	0.36

- Does the data indicate that wax 1 is more effective than wax 2? Test at a 5% level of significance.

Inference on the Mean of Two Population, Variances Known

Definition

If $\bar{x}_1$ and $\bar{x}_2$ are the means of independent random samples of sizes n_1 and n_2 from two independent normal populations with known variances σ_1^2 and σ_2^2 , respectively, a **100(1 - α)% confidence interval for $\mu_1 - \mu_2$** is

$$\bar{x}_1 - \bar{x}_2 - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq \bar{x}_1 - \bar{x}_2 + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of the standard normal distribution.



Inference on the Mean of Two Population, Variances Known

- Choice of Sample Size

$$n = \left(\frac{z_{\alpha/2}}{E} \right)^2 (\sigma_1^2 + \sigma_2^2)$$

- One-Sided Confidence Bounds

- Upper Confidence Bound

$$\mu_1 - \mu_2 \leq \bar{x}_1 - \bar{x}_2 + z_{\alpha} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

- Lower Confidence Bound

$$\bar{x}_1 - \bar{x}_2 - z_{\alpha} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \leq \mu_1 - \mu_2$$

Decision Making for Two Samples



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Inference on the Mean of Two Population, Variances Unknown

- Hypotheses Tests for a Difference in Means, Variances Unknown
- Case 1:

$$\sigma_1^2 = \sigma_2^2 = \sigma^2$$

We wish to test:

$$H_0: \mu_1 - \mu_2 = \Delta_0$$

$$H_1: \mu_1 - \mu_2 \neq \Delta_0$$

The pooled estimator of σ^2 :

The **pooled estimator** of σ^2 , denoted by S_p^2 , is defined by

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

Inference on the Mean of Two Population, Variances Unknown

- Case 1:

$$\sigma_1^2 = \sigma_2^2 = \sigma^2$$

Given the assumptions of this section, the quantity

$$T = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

has a t distribution with $n_1 + n_2 - 2$ degrees of freedom.

Inference on the Mean of Two Population, Variances Unknown

Definition: The Two-Sample or Pooled t-Test*

Null hypothesis: $H_0: \mu_1 - \mu_2 = \Delta_0$

Test statistic:
$$T_0 = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Alternative Hypothesis

$$H_1: \mu_1 - \mu_2 \neq \Delta_0$$

$$H_1: \mu_1 - \mu_2 > \Delta_0$$

$$H_1: \mu_1 - \mu_2 < \Delta_0$$

P-Value

Probability above $|t_0|$ and
probability below $-|t_0|$

Probability above t_0

Probability below t_0

Rejection Criterion for Fixed-Level Tests

$$t_0 > t_{\alpha/2, n_1 + n_2 - 2} \text{ or}$$

$$t_0 < -t_{\alpha/2, n_1 + n_2 - 2}$$

$$t_0 > t_{\alpha, n_1 + n_2 - 2}$$

$$t_0 < -t_{\alpha, n_1 + n_2 - 2}$$

Exercise: Yield from a Catalyst

Two catalysts are being analyzed to determine how they affect the mean yield of a chemical process. Specifically, catalyst 1 is currently in use, but catalyst 2 is acceptable. Since catalyst 2 is cheaper, it should be adopted, providing it does not change the process yield. A test is run in the pilot plant and results in the data shown in table. Is there any difference between the mean yields? Use $\alpha = 0.05$, and assume equal variances.



Observation Number	Catalyst 1	Catalyst 2
1	91.50	89.19
2	94.18	90.95
3	92.18	90.46
4	95.39	93.21
5	91.79	97.19
6	89.07	97.04
7	94.72	91.07
8	89.21	92.75
$\bar{x}_1 = 92.255$		$\bar{x}_2 = 92.733$
$s_1 = 2.39$		$s_2 = 2.98$

Example: Yield from a Catalyst

The solution using the seven-step hypothesis-testing procedure is as follows:

1. **Parameter of interest:** The parameters of interest are μ_1 and μ_2 , the mean process yield using catalysts 1 and 2, respectively, and we want to know if $\mu_1 - \mu_2 = 0$.
2. **Null hypothesis:** $H_0: \mu_1 - \mu_2 = 0$, or $H_0: \mu_1 = \mu_2$
3. **Alternative hypothesis:** $H_1: \mu_1 \neq \mu_2$
4. **Test statistic:** The test statistic is

$$t_0 = \frac{\bar{x}_1 - \bar{x}_2 - 0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

5. **Reject H_0 if:** Reject H_0 if the P -value is less than 0.05.

Example: Yield from a Catalyst

6. **Computations:** From Table 10-1 we have $\bar{x}_1 = 92.255$, $s_1 = 2.39$, $n_1 = 8$, $\bar{x}_2 = 92.733$, $s_2 = 2.98$, and $n_2 = 8$. Therefore

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(7)(2.39)^2 + 7(2.98)^2}{8 + 8 - 2} = 7.30$$

$$s_p = \sqrt{7.30} = 2.70$$

and

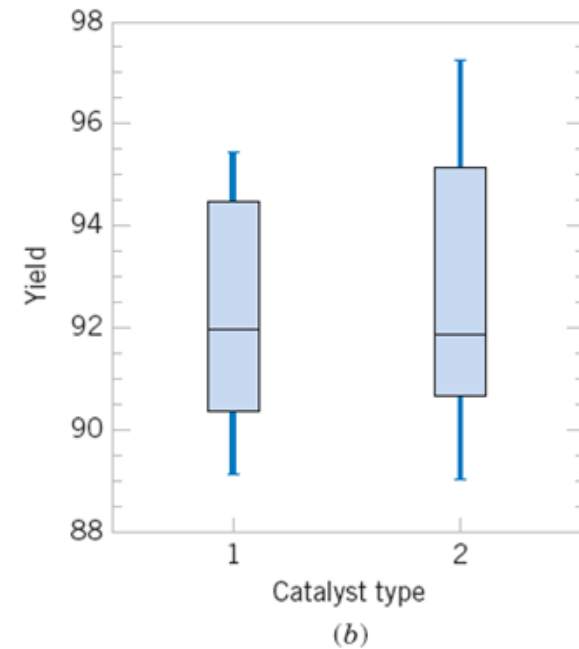
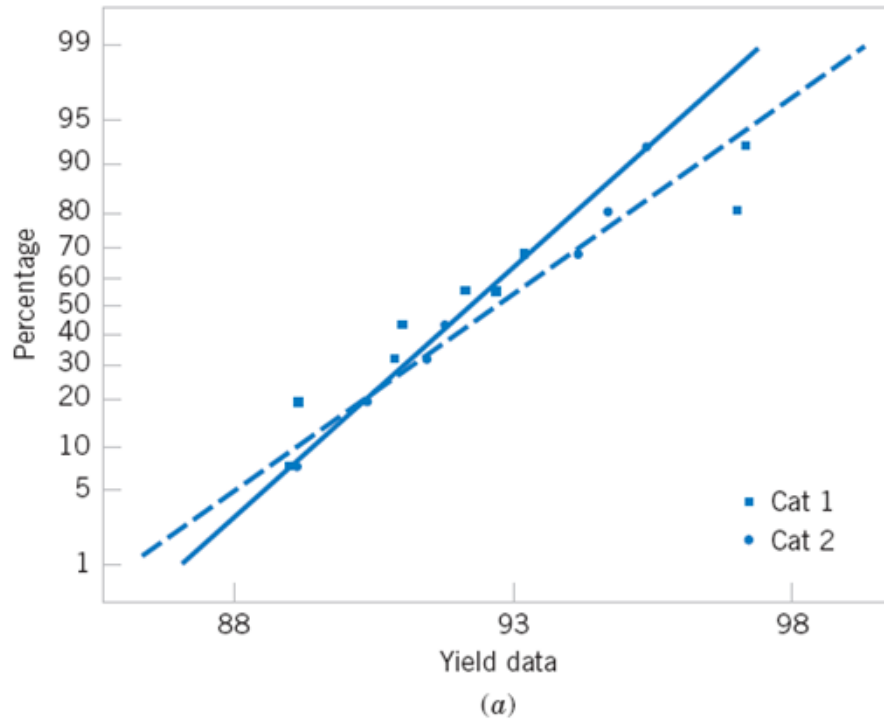
$$t_0 = \frac{\bar{x}_1 - \bar{x}_2}{2.70\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{92.255 - 92.733}{2.70\sqrt{\frac{1}{8} + \frac{1}{8}}} = -0.35$$

7. **Conclusions:** Since -2.145 , from Appendix Table V we find that $t_{0.40,14} = 0.258$ and $t_{0.25,14} = 0.692$. Therefore,

since $0.258 < 0.35 < 0.692$, we conclude that lower and upper bounds on the P -value are $0.50 < P < 0.80$. Therefore, since the P -value exceeds $\alpha = 0.05$, the null hypothesis cannot be rejected.

Practical Interpretation: At the 0.05 level of significance, we do not have strong evidence to conclude that catalyst 2 results in a mean yield that differs from the mean yield when catalyst 1 is used.

Example: Yield from a Catalyst



Normal probability plot and comparative box plot for the catalyst yield data.

(a) Normal probability plot, (b) Box plots.

Inference on the Mean of Two Population, Variances Unknown

- Hypotheses Tests for a Difference in Means, Variances Unknown
- Case 2:

$$\sigma_1^2 \neq \sigma_2^2$$

If $H_0: \mu_1 - \mu_2 = \Delta_0$ is true, the statistic

$$T_0^* = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

is distributed approximately as t with degrees of freedom given by

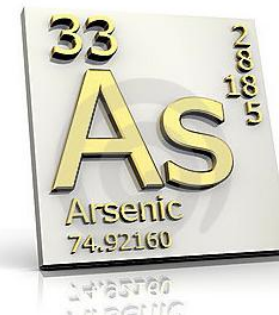
$$v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

Exercise: Arsenic in Drinking Water

Arsenic concentration in public drinking water supplies is a potential health risk. An article in the *Arizona Republic* (May 27, 2001) reported drinking water arsenic concentrations in parts per billion (ppb) for 10 metropolitan Phoenix communities and 10 communities in rural Arizona. The data follow:

$$\alpha = 0.05$$

Metro Phoenix $(\bar{x}_1 = 12.5, s_1 = 7.63)$	Rural Arizona $(\bar{x}_2 = 27.5, s_2 = 15.3)$
Phoenix, 3	Rimrock, 48
Chandler, 7	Goodyear, 44
Gilbert, 25	New River, 40
Glendale, 10	Apache Junction, 38
Mesa, 15	Buckeye, 33
Paradise Valley, 6	Nogales, 21
Peoria, 12	Black Canyon City, 20
Scottsdale, 25	Sedona, 12
Tempe, 15	Payson, 1
Sun City, 7	Casa Grande, 18



Example: Arsenic in Drinking Water

We wish to determine if there is any difference in mean arsenic concentrations between metropolitan Phoenix communities and communities in rural Arizona. Figure 10-3 shows a normal probability plot for the two samples of arsenic concentration. The assumption of normality appears quite reasonable, but since the slopes of the two straight lines are very different, it is unlikely that the population variances are the same.

Applying the seven-step procedure gives the following:

1. **Parameter of interest:** The parameters of interest are the mean arsenic concentrations for the two geographic regions, say, μ_1 and μ_2 , and we are interested in determining whether $\mu_1 - \mu_2 = 0$.
2. **Non hypothesis:** $H_0: \mu_1 - \mu_2 = 0$, or $H_0: \mu_1 = \mu_2$
3. **Alternative hypothesis:** $H_1: \mu_1 \neq \mu_2$
4. **Test statistic:** The test statistic is

$$t_0^* = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Example: Arsenic in Drinking Water

5. The degrees of freedom on t_0^* are found from

$$\begin{aligned} v &= \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} \\ &= \frac{\left[\frac{(7.63)^2}{10} + \frac{(15.3)^2}{10}\right]^2}{\frac{[(7.63)^2/10]^2}{9} + \frac{[(15.3)^2/10]^2}{9}} = 13.2 \approx 13 \end{aligned}$$

Therefore, using $\alpha = 0.05$ and a fixed-significance-level test, we would reject $H_0: \mu_1 = \mu_2$ if $t_0^* > t_{0.025,13} = 2.160$ or if $t_0^* < -t_{0.025,13} = -2.160$.

Example: Arsenic in Drinking Water

6. **Computations:** Using the sample data we find

$$t_0^* = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{12.5 - 27.5}{\sqrt{\frac{(7.63)^2}{10} + \frac{(15.3)^2}{10}}} = -2.77$$

7. **Conclusions:** Because $t_0^* = -2.77 < t_{0.025,13} = -2.160$, we reject the null hypothesis.

Practical Interpretation: There is strong evidence to conclude that mean arsenic concentration in the drinking water in rural Arizona is different from the mean arsenic concentration in metropolitan Phoenix drinking water. Furthermore, the mean arsenic concentration is higher in rural Arizona communities. The P -value for this test is approximately $P = 0.016$.

Inference on the Mean of Two Population, Variances Unknown

- Confidence Interval on the Difference in Means, Variance Unknown
- Case 1:

$$\sigma_1^2 = \sigma_2^2 = \sigma^2$$

If $\bar{x}_1, \bar{x}_2, s_1^2$, and s_2^2 are the sample means and variances of two random samples of sizes n_1 and n_2 , respectively, from two independent normal populations with unknown but equal variances, then a $100(1 - \alpha)\%$ confidence interval on the difference in means $\mu_1 - \mu_2$ is

$$\begin{aligned} \bar{x}_1 - \bar{x}_2 - t_{\alpha/2, n_1+n_2-2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \\ \leq \mu_1 - \mu_2 \leq \bar{x}_1 - \bar{x}_2 + t_{\alpha/2, n_1+n_2-2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \end{aligned}$$

where $s_p = \sqrt{[(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2]/(n_1 + n_2 - 2)}$ is the pooled estimate of the common population standard deviation, and $t_{\alpha/2, n_1+n_2-2}$ is the upper $\alpha/2$ percentage point of the t distribution with $n_1 + n_2 - 2$ degrees of freedom.

Inference on the Mean of Two Population, Variances Unknown

- Confidence Interval on the Difference in Means, Variance Unknown
- Case 2:

$$\sigma_1^2 \neq \sigma_2^2$$

If $\bar{x}_1$, $\bar{x}_2$, s_1^2 , and s_2^2 are the means and variances of two random samples of sizes n_1 and n_2 , respectively, from two independent normal populations with unknown and unequal variances, an approximate $100(1 - \alpha)\%$ confidence interval on the difference in means $\mu_1 - \mu_2$ is

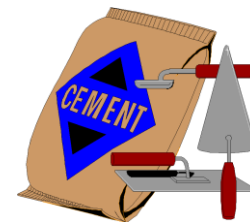
$$\bar{x}_1 - \bar{x}_2 - t_{\alpha/2, v} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq \bar{x}_1 - \bar{x}_2 + t_{\alpha/2, v} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

where v is given by Equation 10-16 and $t_{\alpha/2, v}$ is the upper $\alpha/2$ percentage point of the t distribution with v degrees of freedom.

Exercise: Cement Hydration

An article in the journal *Hazardous Waste and Hazardous Materials* (Vol. 6, 1989) reported the results of an analysis of the weight of calcium in standard cement and cement doped with lead. Reduced levels of calcium would indicate that the hydration mechanism in the cement is blocked and would allow water to attack various locations in the cement structure. Ten samples of standard cement had an average weight percent calcium of $\bar{x}_1 = 90.0$, with a sample standard deviation of $s_1 = 5.0$, while 15 samples of the lead-doped cement had an average weight percent calcium of $\bar{x}_2 = 87.0$, with a sample standard deviation of $s_2 = 4.0$.

We will assume that weight percent calcium is normally distributed and find a 95% confidence interval on the difference in means, $\mu_1 - \mu_2$, for the two types of cement. Furthermore, we will assume that both normal populations have the same standard deviation.



Example: Cement Hydration

The pooled estimate of the common standard deviation is found as follows:

$$\begin{aligned}s_p^2 &= \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} \\&= \frac{9(5.0)^2 + 14(4.0)^2}{10 + 15 - 2} \\&= 19.52\end{aligned}$$

standard deviation.

Therefore, the pooled standard deviation estimate is $s_p = \sqrt{19.52} = 4.4$. The 95% confidence interval is found using Equation 10-19:

$$\begin{aligned}\bar{x}_1 - \bar{x}_2 - t_{0.025,23} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} &\leq \mu_1 - \mu_2 \\&\leq \bar{x}_1 - \bar{x}_2 + t_{0.025,23} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}\end{aligned}$$

or upon substituting the sample values and using $t_{0.025,23} = 2.069$,

$$\begin{aligned}90.0 - 87.0 - 2.069(4.4) \sqrt{\frac{1}{10} + \frac{1}{15}} &\leq \mu_1 - \mu_2 \\&\leq 90.0 - 87.0 + 2.069(4.4) \sqrt{\frac{1}{10} + \frac{1}{15}}\end{aligned}$$

Example: Cement Hydration

which reduces to

$$-0.72 \leq \mu_1 - \mu_2 \leq 6.72$$

Practical Interpretation: Notice that the 95% confidence interval includes zero; therefore, at this level of confidence we cannot conclude that there is a difference in the means. Put another way, there is no evidence that doping the cement with lead affected the mean weight percent of calcium; therefore, we cannot claim that the presence of lead affects this aspect of the hydration mechanism at the 95% level of confidence.

Decision Making for Two Samples



- Inference on the Mean of Two Population, Variances Known
- Inference on the Mean of Two Population, Variances Unknown
- **The Paired t-Test**
- Inference on the Ratio of Variances of Two Normal Populations
- Inference on Two Population Proportions

Paired t-Test

- A special case of the two-sample t -tests occurs when the observations on the two populations of interest are collected in **pairs**.
- Each pair of observations, say (X_{1j}, X_{2j}) , is taken under homogeneous conditions, but these conditions may change from one pair to another.
- The test procedure consists of analyzing the **differences** between hardness readings on each specimen.

Paired t-Test

- The Paired t-Test

Null hypothesis: $H_0: \mu_D = \Delta_0$

Test statistic: $T_0 = \frac{\bar{D} - \Delta_0}{S_D/\sqrt{n}}$

**Alternative
Hypothesis**

***P*-Value**

**Rejection Region
for Fixed-Level Tests**

$H_1: \mu_D \neq \Delta_0$

Sum of the probability above t_0 and
the probability below $-t_0$,

$t_0 > t_{\alpha/2, n-1}$ or $t_0 < -t_{\alpha/2, n-1}$

$H_1: \mu_D > \Delta_0$

Probability above t_0

$t_0 > t_{\alpha, n-1}$

$H_1: \mu_D < \Delta_0$

Probability below t_0

$t_0 < -t_{\alpha, n-1}$

Paired t-Test

- Paired Versus Unpaired Comparisons

So how do we decide to conduct the experiment? Should we pair the observations or not? Although there is no general answer to this question, we can give some guidelines based on the above discussion.

1. If the experimental units are relatively homogeneous (small σ) and the correlation within pairs is small, the gain in precision attributable to pairing will be offset by the loss of degrees of freedom, so an independent-sample experiment should be used.
2. If the experimental units are relatively heterogeneous (large σ) and there is large positive correlation within pairs, the paired experiment should be used. Typically, this case occurs when the experimental units are the *same* for both treatments.

Paired t-Test

- A Confidence Interval for μ_D

If $\bar{d}$ and s_D are the sample mean and standard deviation of the difference of n random pairs of normally distributed measurements, a $100(1 - \alpha)\%$ confidence interval on the difference in means $\mu_D = \mu_1 - \mu_2$ is

$$\bar{d} - t_{\alpha/2, n-1} s_D / \sqrt{n} \leq \mu_D \leq \bar{d} + t_{\alpha/2, n-1} s_D / \sqrt{n}$$

where $t_{\alpha/2, n-1}$ is the upper $\alpha/2\%$ point of the t distribution with $n - 1$ degrees of freedom.

Exercise: Shear Strength of Steel Girders

An article in the *Journal of Strain Analysis* (Vol. 18, No. 2 1983) reports a comparison of several methods for predicting the shear strength for steel plate girders. Data for two of these methods, the Karlsruhe and Lehigh procedures, when applied to nine specific girders, are shown in the Table. We wish to determine whether there is any difference (on the average) between the two methods.



$$\alpha = 0.001$$

Girder	Karlsruhe Method	Lehigh Method	Difference d_j
S1/1	1.186	1.061	0.125
S2/1	1.151	0.992	0.159
S3/1	1.322	1.063	0.259
S4/1	1.339	1.062	0.277
S5/1	1.200	1.065	0.135
S2/1	1.402	1.178	0.224
S2/2	1.365	1.037	0.328
S2/3	1.537	1.086	0.451
S2/4	1.559	1.052	0.507

Decision Making for Two Samples



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Inferences on the Variances of Two Normal Populations

The F Distribution

- We wish to test the hypotheses:

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

- The development of a test procedure for these hypotheses requires a new probability distribution, the F distribution.

Inferences on the Variances of Two Normal Populations

- The F Distribution

Let W and Y be independent chi-square random variables with u and v degrees of freedom, respectively. Then the ratio

$$F = \frac{W/u}{Y/v}$$

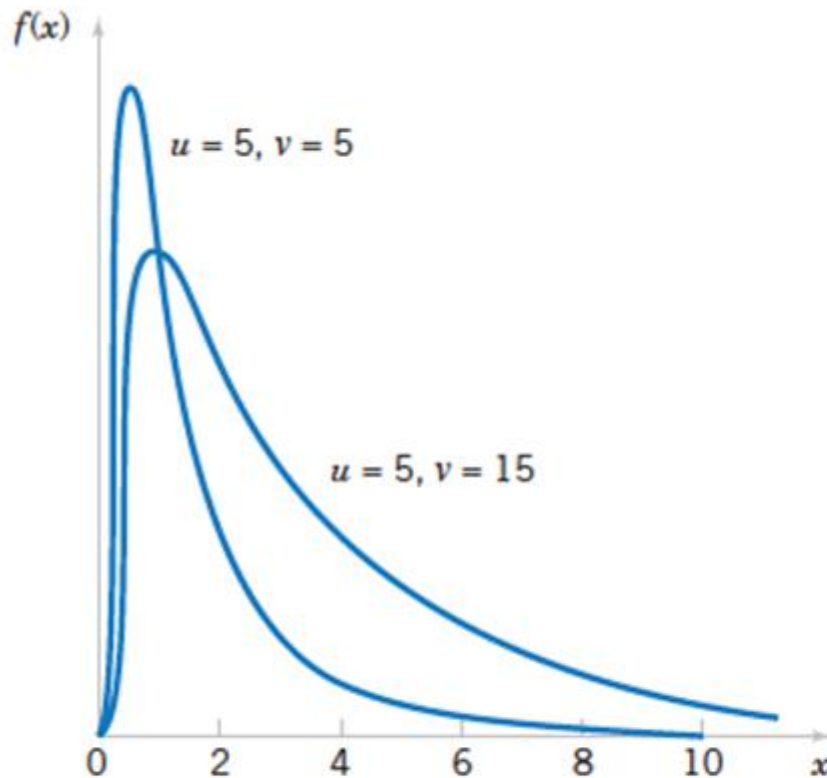
has the probability density function

$$f(x) = \frac{\Gamma\left(\frac{u+v}{2}\right)\left(\frac{u}{v}\right)^{u/2} x^{(u/2)-1}}{\Gamma\left(\frac{u}{2}\right)\Gamma\left(\frac{v}{2}\right)\left[\left(\frac{u}{v}\right)x + 1\right]^{(u+v)/2}}, \quad 0 < x < \infty$$

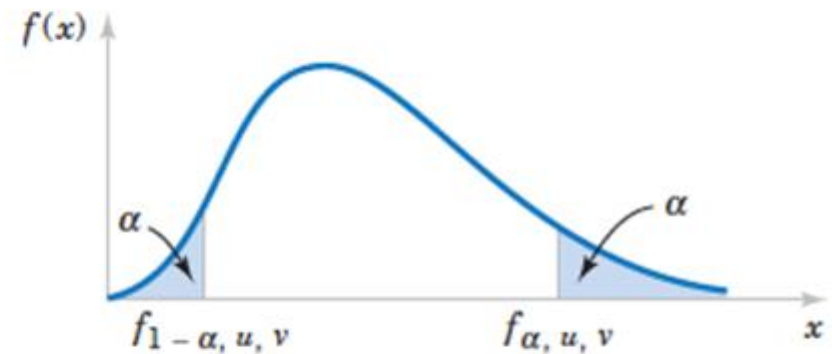
and is said to follow the F distribution with u degrees of freedom in the numerator and v degrees of freedom in the denominator. It is usually abbreviated as $F_{u,v}$.

Inferences on the Variances of Two Normal Populations

- The F Distribution



Probability density functions of two F distributions.



Upper and lower percentage points of the F distribution.

Inferences on the Variances of Two Normal Populations

The F Distribution

- The lower-tail percentage points $f_{\alpha-1,u,v}$ can be found as follows.

$$f_{1-\alpha,u,v} = \frac{1}{f_{\alpha,v,u}}$$

Inferences on the Variances of Two Normal Populations

Hypothesis Tests on the Ratio of Two Variances

Let $X_{11}, X_{12}, \dots, X_{1n_1}$ be a random sample from a normal population with mean μ_1 and variance σ_1^2 , and let $X_{21}, X_{22}, \dots, X_{2n_2}$ be a random sample from a second normal population with mean μ_2 and variance σ_2^2 . Assume that both normal populations are independent. Let S_1^2 and S_2^2 be the sample variances. Then the ratio

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$$

has an F distribution with $n_1 - 1$ numerator degrees of freedom and $n_2 - 1$ denominator degrees of freedom.

Inferences on the Variances of Two Normal Populations

Hypothesis Tests on the Ratio of Two Variances

Null hypothesis: $H_0: \sigma_1^2 = \sigma_2^2$

Test statistic: $F_0 = \frac{S_1^2}{S_2^2}$

Alternative Hypotheses

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

$$H_1: \sigma_1^2 > \sigma_2^2$$

$$H_1: \sigma_1^2 < \sigma_2^2$$

Rejection Criterion

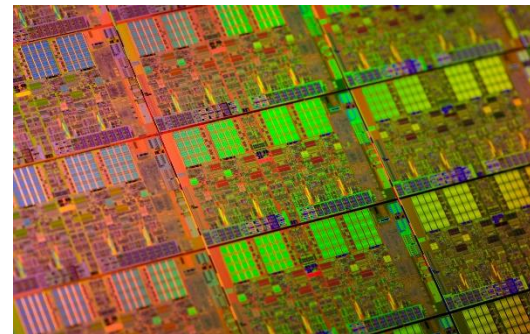
$$f_0 > f_{\alpha/2, n_1-1, n_2-1} \text{ or } f_0 < f_{1-\alpha/2, n_1-1, n_2-1}$$

$$f_0 > f_{\alpha, n_1-1, n_2-1}$$

$$f_0 < f_{1-\alpha, n_1-1, n_2-1}$$

Exercise: Semiconductor Etch Variability

Oxide layers on semiconductor wafers are etched in a mixture of gases to achieve the proper thickness. The variability in the thickness of these oxide layers is a critical characteristic of the wafer, and low variability is desirable for subsequent processing steps. Two different mixtures of gases are being studied to determine whether one is superior in reducing the variability of the oxide thickness. Sixteen wafers are etched in each gas. The sample standard deviations of oxide thickness are $s_1 = 1.96$ angstroms and $s_2 = 2.13$ angstroms, respectively. Is there any evidence to indicate that either gas is preferable? Use a fixed-level test with $\alpha = 0.05$.



Inferences on the Variances of Two Normal Populations

Confidence Interval on the Ratio of Two Variances

If s_1^2 and s_2^2 are the sample variances of random samples of sizes n_1 and n_2 , respectively, from two independent normal populations with unknown variances σ_1^2 and σ_2^2 , then a $100(1 - \alpha)\%$ confidence interval on the ratio σ_1^2/σ_2^2 is

$$\frac{s_1^2}{s_2^2} f_{1-\alpha/2, n_2-1, n_1-1} \leq \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{s_1^2}{s_2^2} f_{\alpha/2, n_2-1, n_1-1}$$

where $f_{\alpha/2, n_2-1, n_1-1}$ and $f_{1-\alpha/2, n_2-1, n_1-1}$ are the upper and lower $\alpha/2$ percentage points of the F distribution with $n_2 - 1$ numerator and $n_1 - 1$ denominator degrees of freedom, respectively. A confidence interval on the ratio of the standard deviations can be obtained by taking square roots in Equation 10-33.

Exercise: Confidence Interval on the Ratio of Two Variances

Surface Finish for Titanium Alloy

A company manufactures impellers for use in jet-turbine engines. One of the operations involves grinding a particular surface finish on a titanium alloy component. Two different grinding processes can be used, and both processes can produce parts at identical mean surface roughness. The manufacturing engineer would like to select the process having the least variability in surface roughness. A random sample of $n_1 = 11$ parts from the first process results in a sample standard deviation $s_1 = 5.1$ microinches, and a random sample of $n_2 = 16$ parts from the second process results in a sample standard deviation of $s_2 = 4.7$ microinches. We will find a 90% confidence interval on the ratio of the two standard deviations, σ_1/σ_2 .

Decision Making for Two Samples



- Inference on the Mean of Two Population, Variances Known
- Inference on the Mean of Two Population, Variances Unknown
- The Paired t-Test
- Inference on the Ratio of Variances of Two Normal Populations
- Inference on Two Population Proportions

Inference on Two Population Proportions

Large-Sample Test on the Difference in Population Proportions

- We wish to test the hypotheses:

$$H_0: p_1 = p_2$$

$$H_1: p_1 \neq p_2$$

- The following test statistic is distributed approximately as standard normal and is the basis of the test:

$$Z = \frac{\hat{p}_1 - \hat{p}_2 - (p_1 - p_2)}{\sqrt{\frac{p_1(1 - p_1)}{n_1} + \frac{p_2(1 - p_2)}{n_2}}}$$

Inference on Two Population Proportions

Large-Sample Test on the Difference in Population Proportions

Null hypothesis: $H_0: p_1 = p_2$

Test statistic:
$$Z_0 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Alternative Hypothesis	P-Value	Rejection Criterion for Fixed-Level Tests
$H_1: p_1 \neq p_2$	Probability above $ z_0 $ and probability below $- z_0 $. $P = 2[1 - \Phi(z_0)]$	$z_0 > z_{\alpha/2}$ or $z_0 < -z_{\alpha/2}$
$H_1: p_1 > p_2$	Probability above z_0 . $P = 1 - \Phi(z_0)$	$z_0 > z_\alpha$
$H_1: p_1 < p_2$	Probability below z_0 . $P = \Phi(z_0)$	$z_0 < -z_\alpha$

Exercise: St. John's Wort

Extracts of St. John's Wort are widely used to treat depression. An article in the April 18, 2001, issue of the *Journal of the American Medical Association* ("Effectiveness of St. John's Wort on Major Depression: A Randomized Controlled Trial") compared the efficacy of a standard extract of St. John's Wort with a placebo in 200 outpatients diagnosed with major depression. Patients were randomly assigned to two groups; one group received the St. John's Wort, and the other received the placebo. After eight weeks, 19 of the placebo-treated patients showed improvement, whereas 27 of those treated with St. John's Wort improved. Is there any reason to believe that St. John's Wort is effective in treating major depression? Use $\alpha = 0.05$.



Inference on Two Population Proportions

Confidence Interval on the Difference in the Population Proportions

If $\hat{p}_1$ and $\hat{p}_2$ are the sample proportions of observations in two independent random samples of sizes n_1 and n_2 that belong to a class of interest, an **approximate two-sided $100(1 - \alpha)\%$ confidence interval on the difference in the true proportions $p_1 - p_2$** is

$$\hat{p}_1 - \hat{p}_2 - z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} \leq p_1 - p_2 \leq \hat{p}_1 - \hat{p}_2 + z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of the standard normal distribution.

Example

Defective Bearings

Consider the process of manufacturing crankshaft bearings.

Suppose that a modification is made in the surface finishing process and that, subsequently, a second random sample of 85 bearings is obtained. The number of defective bearings in this second sample is 8. Therefore, since $n_1 = 85$, $\hat{p}_1 = 0.12$, $n_2 = 85$, and $\hat{p}_2 = 8/85 = 0.09$, we can obtain an approximate 95% confidence interval on the difference in the proportion of defective bearings produced under the two processes as follows:

$$\begin{aligned} \hat{p}_1 - \hat{p}_2 - z_{0.025} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} \\ \leq p_1 - p_2 \leq \hat{p}_1 - \hat{p}_2 + z_{0.025} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} \end{aligned}$$